PO Box 967 | Bozeman, Montana 59771
tel 833.685.2872 | fax 833.395.2872 | www.LightCon.com

Quote #: Q261016-AS1

October 19, 2026

Dr. Oluwatobi Busari
Morgan State University, Clarence M. Mitchell Jr. School of Engineering
1700 East Cold Spring Lane, Baltimore, MD 21251
(443) 885-4237; oluwatobi.busari@morgan.edu

Dear Dr. Oluwatobi Busari,
Following is the quotation for the CARBIDE-40W-0.4mJ_ORPHEUS-F tunable laser system configured for spectroscopy. Key features and specifications include:

- CARBIDE pump laser has been engineered for applications where reliability is crucial, it represents 'best in class' stability and a robust design. The CPA design with computer-controlled stretcher/ compressor allows the user to control the pulse duration from a computer. Tunability of the laser output parameters allows CARBIDE system to cover applications normally requiring different classes of lasers.
- Orpheus-F is a hybrid OPA combining the best features of collinear and non-collinear OPA and offering computer controlled wavelength tuning as well as operation in short-pulse (< 100 fs) mode.
- The entire system has been designed to allow customers to use the laser as a tool, maintaining focus on their work. It is thoroughly tested at production site and debugged there to ensure best performance at installation.
- The system is equipped with an extensive software package, which ensures smooth operation as well as remote system monitoring and troubleshooting.

We have designed the system to be flexible, and are happy to work with you over time to update or reconfigure the system to accommodate evolutions in your work.

We appreciate the opportunity to work with your group, please let me know if you have any questions.

Best Regards

Andrey Senin, Ph.D.
Light Conversion-USA
Andrey@LightCon-USA.com
(833) 685-2872 x118

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1700 East Cold Spring Lane, Baltimore, MD 21251; (443) 885-4237; oluwatobi.busari@morgan.edu

Item	Description	Price, USD
Carbide-40W-0.4mJ-1MHz	CARBIDE-40W-0.4mJ DPSS fs pump laser • ~1030 nm Fundamental Output • Output power: 40 W max; Max output energy: 0.4 mJ/pulse • < 250 fs fundamental pulse duration; Rep rate: 100 kHz - 1000 kHz • Includes integral pulse picker, which may be used to effectively divide PRR • Includes a laser control laptop	$ 190,640
ORPHEUS-F	ORPHEUS-F broad bandwidth optical parametric amplifier • 650-900 nm (signal) and 1200-2500 nm (idler) • Input energy (power & rep rate): *400uJ (40W @ 100kHz) - pls. confirm* • Pulse duration after compression <55fs @ 800-900 nm, <70fs @ 650-800 nm, <100fs @ 1200-2000 nm • Compressor transmission: >65 % @ 650-900 nm, >80 % @ 1200-2000 nm	$ 69,240
	Equipment Subtotal:	***$ 259,880***
	10% discount for optics, crystals, lasershields & mechanics within up to 3 months of purchase: Opto-Mechanics: http://www.standa.lt/support/download Laser components: http://eksmaoptics.com/catalogues/	Included
	Delivery (by air freight, to a loading dock)	$ 3,300
	Installation & Training (includes airfare, arrival and expenses)	$ 7,900
	TOTAL (not including an import fee and sales tax):	**$ 271,080**

Option 1	Warranty Extension for the 2nd year for the 40W pumped system • Must be purchased prior to expiration of the 1st year warranty	$ 16,320
Option 2	Warranty Extension for the 3d year for the 40W pumped system • Must be purchased prior to expiration of the 2nd year warranty	$ 23,940
Option 3	Upgrade to 80W max - 0.4mJ max Carbide pump laser • Output power: 80 W max (at 200 kHz to 2000 kHz) • Max output energy: 0.4 mJ/pulse (at up to 200 kHz)	$ 44,940
Option 4	Warranty Extension for the 2nd year for the 80W pumped system • Must be purchased prior to expiration of the 1st year warranty	$ 16,320
Option 5	Warranty Extension for the 3d year for the 80W pumped system • Must be purchased prior to expiration of the 2nd year warranty	$ 25,740

NOTES:

1. Prices quoted include US educational/government discount.
2. Local sales tax will be added unless exemption certificate is provided.
3. Pursuant to the tariff policies enacted by the White House, an import duty applies to virtually all imports from the European Union (including Lithuania). Currently the rate is 10%, but is subject to change. We are forced to add these import fees to the order total, but due to the current volatility we cannot predict in advance what fee will apply at the time of shipment. Please note that the quoted prices do NOT include the import fee. We hope that these punitive tariffs will be reduced or repealed, so we can return to normal operations. In the meantime, we deeply regret having to impose this fee, and we appreciate your understanding..

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Terms & Conditions

- **Price:** US Dollars
- **Terms:** Incoterms ***DDP***
- **Payment:** Net 30, payment by check or wire transfer
- **Delivery:** 6-8 months ARO
- **Quotation valid:** 30 days

CLASS IV LASER SYSTEM: User is responsible for compliance with all local codes and regulations.

Product Warranty: Seller warrants to Buyer that the Products shall be substantially free from defects in materials and workmanship under normal use and service for one (1) year from the date of shipment from the factory.

Buyer's exclusive remedy and Seller's sole liability for any breach of the foregoing warranty shall be for Seller, at Buyer's request within the warranty period by written notice specifying the defect, and at Seller's sole option, to repair or replace the defective Product or refund any amounts paid for the defective Product.

For on-site warranty service, labor and daily expenses are paid by LC-USA. For on-site service, typical response time is 2-4 weeks. If immediate service is needed, buyer agrees to pay for airfare.

These remedies are available only if Seller's examination discloses to Seller's satisfaction that such defects actually exist and were not caused by Buyer's misuse, abuse, unauthorized modifications or disassembly, neglect, attempts to repair, or by accident, fire, third party materials or other hazard. It is buyer's responsibility to perform periodic maintenance including periodically inspecting OPA optics, cleaning as-needed. Repair or replacement of a part does not extend the warranty period beyond the initial warranty period, which commences with the date of shipment.

Seller makes no representation or warranty that the Products supplied hereunder comply with any local laws or ordinances, and Buyer has the responsibility for compliance with local laws and ordinances, including obtaining all permits, licenses, authorizations or certificates required by any regulatory body for installation or use of the Products.

Service Warranty: Services shall be performed consistent with generally prevailing professional and industry standards. Buyer's exclusive remedy and Seller's sole liability for any breach of the foregoing warranty shall be the re-performance of the applicable Services, failing which Seller shall refund the portion of fees paid which relate to the specific non-conforming Services.

No Other Warranties: To the maximum extent permitted by applicable law, the Seller and its suppliers disclaim all other warranties, either expressed or implied, including but not limited to implied warranties of merchantability and fitness for a particular purpose, with regard to the software and any related or accompanying written materials. No liability for damages, to the maximum extent permitted.

Laser Safety: Laser safety training is not provided by LC-USA. It is imperative that the customer obtain laser safety training from a qualified Laser Safety Officer, and ensure that lasers are operated in a safe manner and in compliance with applicable regulations. We offer training on use and maintenance of the systems we sell. However, this training does not substitute for laser safety training.

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Technical Specifications

- *See product brochure for detailed specifications,* www.LightCon.com

High Power Femtosecond DPSSL System CARBIDE-40W

- Integrated design
- Yb lasing medium
- All solid state diode pump technology

Fundamental Wavelength	1030 +/- 10 nm
Pump Laser Output Power	40 W maximum average power at 100 kHz to 1 MHz
Pulse energy	400 µJ maximum output energy at up to 100 kHz
Pulse duration :	< 250 fs (assuming Gaussian pulse shape)
Pulse duration adjustment range	250 fs – 10 ps
Repetition rate tunable in the range of	100 kHz – 1 MHz (lower repetition rates available using Pulse Picker)
Pulse selection	Single-shot, Pulse-on-Demand, any base repetition rate division
Output beam quality	$M^2 < 1.2$
Output pulse-to-pulse stability	< 0.5 % nrmsd over 8 hours (under stable environmental conditions)
Power stability	< 0.3 % nrmsd over 100 hours (under stable environmental conditions)
Beam pointing stability	< 20 µrad
Laser settings and functions	PC controllable
Laser diagnostics	available through PC interface

Powering	120 VAC or 208 VAC, single phase, < 2 kW
Safety	Includes interlock and emergency stop connections
Cooling:	Air-cooled or water-cooled chiller is included (choose when ordering)

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High Power Femtosecond DPSSL System CARBIDE-80W (*see option #3*)

- Integrated design
- Yb lasing medium
- All solid state diode pump technology

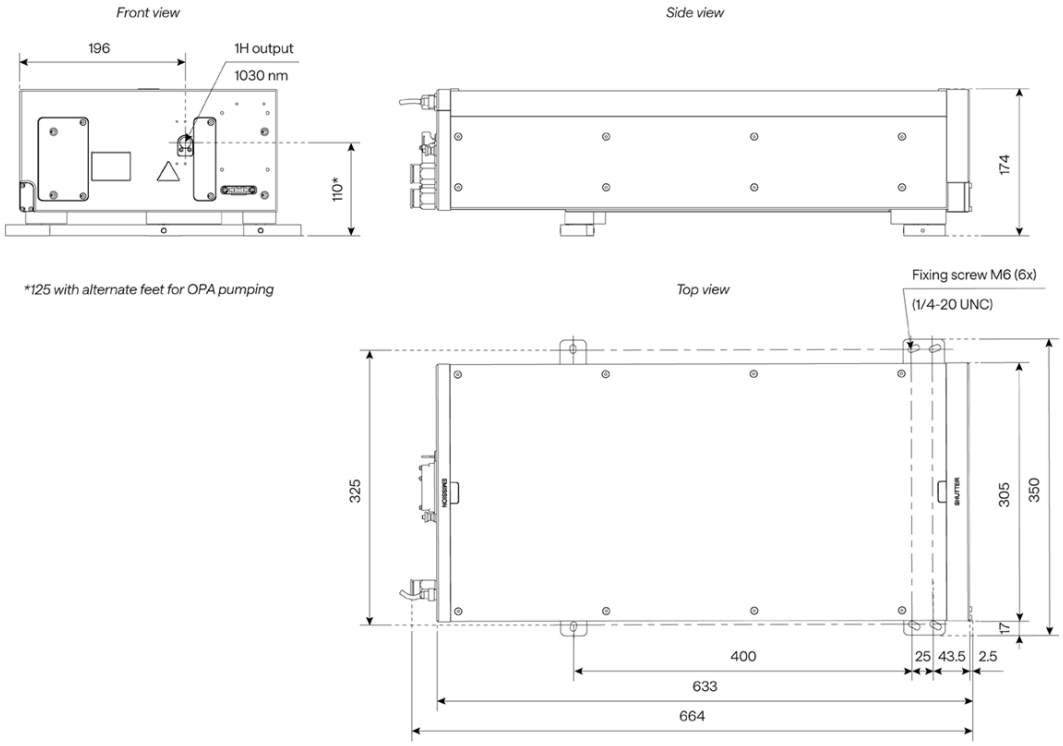
Fundamental Wavelength	1030 +/- 10 nm
Pump Laser Output Power	80 W maximum average power at 200 kHz to 2 MHz
Pulse energy	400 µJ maximum output energy at up to 200 kHz
Pulse duration :	< 250 fs (assuming Gaussian pulse shape)
Pulse duration adjustment range	250 fs – 10 ps
Repetition rate tunable in the range of	200 kHz –2 MHz (lower repetition rates available using Pulse Picker)
Pulse selection	Single-shot, Pulse-on-Demand, any base repetition rate division
Output beam quality	$M^2 < 1.2$
Output pulse-to-pulse stability	< 0.5 % nrmsd over 8 hours (under stable environmental conditions)
Power stability	< 0.3 % nrmsd over 100 hours
Beam pointing stability	< 20 µrad
Laser settings and functions	PC controllable
Laser diagnostics	available through PC interface

Powering	Laser power supply: 120 VAC or 208 VAC, single phase, < 2 kW Chiller requires 208 VAC (NEMA L6-20 plug)
Safety	Includes interlock and emergency stop connections
Cooling:	Air-cooled or water-cooled chiller is included (choose when ordering)

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Carbide-40W/ 80W laser Dimensions (mm)

(Note: available output ports depend on the quoted configuration)





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ORPHEUS-F OPA specifications

Tuning range	650 nm – 900 nm (signal) , 1200 nm – 2500 nm (idler)
Conversion efficiency at peak of tuning curve (signal and idler combined)	>10 %
Long term power stability (8 h)	<2 % nrmsd @ 800 nm
Pulse bandwidth	200 – 750 cm^{-1} @ 650 – 900 nm
Pulse duration after compression	< 55 fs @ 800 – 900 nm < 70 fs @ 650 – 800 nm < 100 fs @ 1200 – 2000 nm Typical: 25 – 70 fs @ 650 – 900 nm 40 – 100 fs @ 1200 – 2000 nm
Compressor transmission	> 65 % @ 650 – 900 nm > 80 % @ 1200 – 2000 nm

Notes: Conversion efficiency is specified as percentage of the input power to the ORPHEUS-F OPA.
Optional compressor includes two prism compressor for signal (external) and bulk compressor for idler (internal).
When operating at rep rate of 1 MHz or above, the following components are considered consumable: amplification & doubling crystals and SH optics
Specifications are guaranteed for a maximum temperature variation of ± 1 ºC and humidity variation of ± 10%.

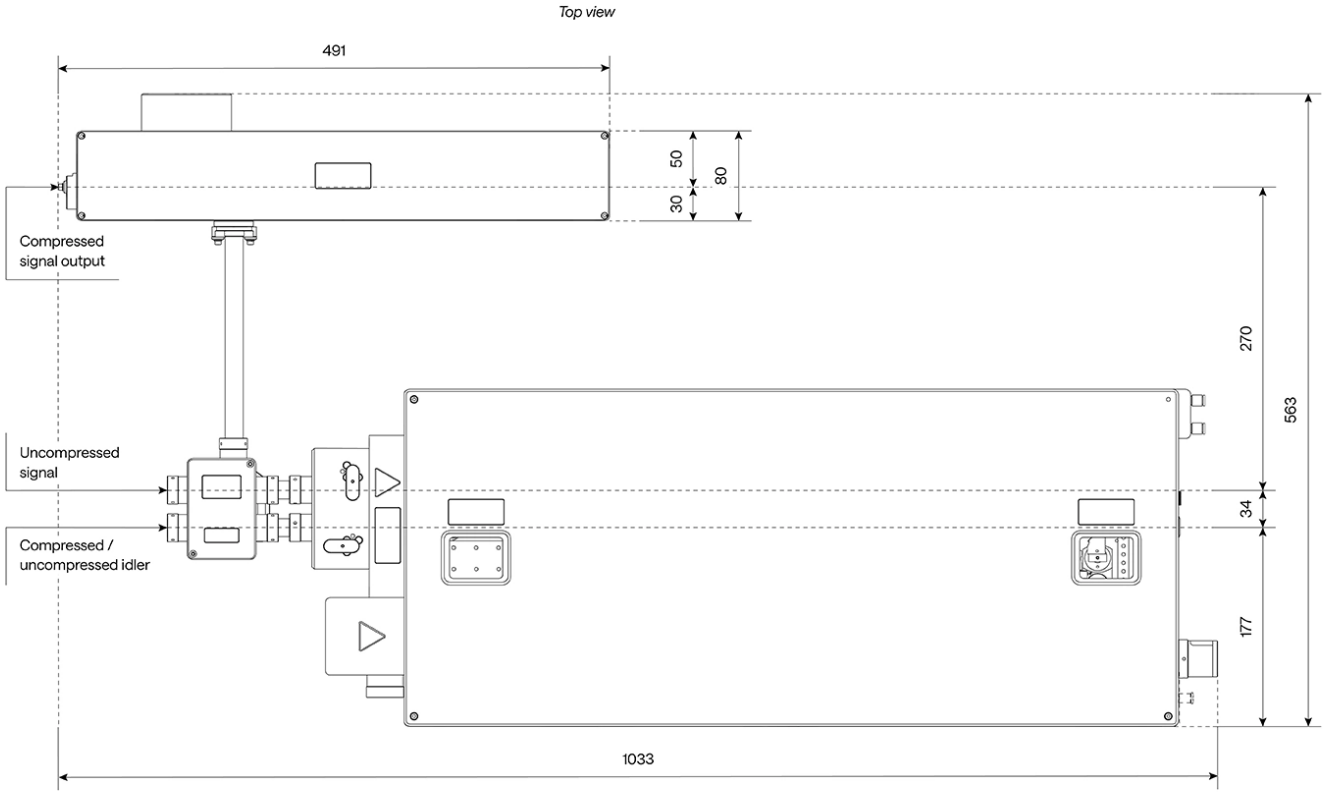
Pump Laser Requirements

Parameter	Value
Wavelength	1030 nm
Pump energy	10 – 400 µJ
Pump power	up to 80 W
Pulse duration (FWHM)	<250 fs

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Orpheus-F Drawing (mm)

(with an external signal compressor shown)

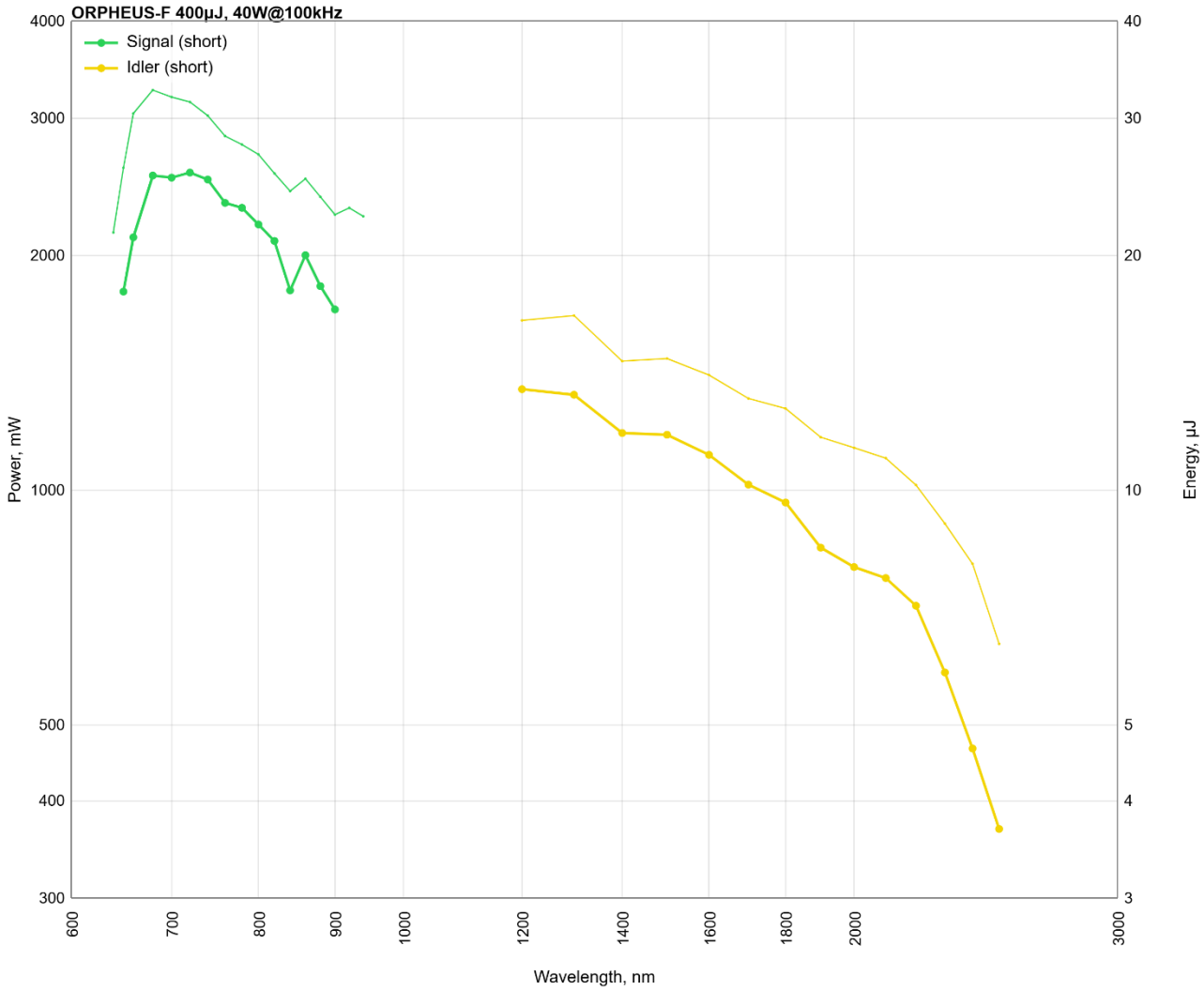




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Sample tuning curve for ORPHEUS-F OPA operating at *40W @ 100kHz*

(Sample tuning curves can be found at http://toolbox.lightcon.com/tools/tuningcurves/
Layout planner tool can be found at http://toolbox.lightcon.com/planner)



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Sample tuning curve for ORPHEUS-F OPA operating at *80W @ 200kHz*

(Sample tuning curves can be found at http://toolbox.lightcon.com/tools/tuningcurves/
Layout planner tool can be found at http://toolbox.lightcon.com/planner)

